

Generic Degree Two in $JC(2)$: Secant Projectors and Galois Rigidity

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Abstract

We study planar polynomial maps of generic degree two through the scheme of pairs with the same image. For

$$F = (P, Q) : \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2,$$

put $A = \mathbb{C}[x_1, x_2]$, let $I_R(F)$ be the collision ideal in the pair ring $S = \mathbb{C}[x_1, x_2, y_1, y_2]$, and let I_{Δ} be the diagonal ideal. Set $C_F = S/I_R(F)$. Their quotient

$$\text{Obs}(F) = I_{\Delta}/I_R(F) \subseteq C_F$$

is the ideal of the diagonal in the collision ring. When F is Keller, a divided-difference secant matrix produces an explicit idempotent $p_F \in C_F$ with $\text{Obs}(F) = C_F p_F$.

Writing $B = \mathbb{C}[P, Q] \subseteq A$, $K = \text{Frac}(B)$, and $L = \text{Frac}(A)$, suppose that the induced extension L/K has degree two. Then the generic collision algebra decomposes compatibly with the diagonal as

$$K \otimes_B C_F \simeq_K L \times L,$$

where diagonal evaluation is first projection. The second factor forces $\text{Obs}(F) \neq 0$; when F is Keller, the secant identity $\text{Obs}(F) = C_F p_F$ then gives $p_F \neq 0$. On the other hand, a separable quadratic extension is Galois, and classical Keller–Galois rigidity makes a hypothetical generic-degree-two Keller map a polynomial automorphism. Then $\text{Obs}(F) = 0$ and $p_F = 0$, a contradiction. Thus no planar Keller map has generic degree two. The collision and secant arguments are formalized in Lean; the classical rigidity theorem is isolated there as the sole external mathematical input to the degree-two exclusion.

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1 Introduction

The complex two-dimensional Jacobian conjecture $JC(2)$ asks whether every polynomial map

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF \in \mathbb{C}^\times,$$

has a polynomial inverse. Such an F is called a *Keller map*. The condition on the Jacobian is local, while polynomial invertibility is global. Collision geometry makes the global issue visible by retaining pairs (x, y) for which $F(x) = F(y)$.

The companion paper [vdV26] develops the dimension-independent collision construction. It associates to a polynomial map two ideals

$$I_R(F) = (F_i(x) - F_i(y)), \quad I_\Delta = (x_i - y_i)$$

inside the coordinate ring of ordered pairs. The containment $I_R(F) \subseteq I_\Delta$ induces the obstruction ideal

$$\text{Obs}(F) = I_\Delta / I_R(F)$$

in the collision ring C_F . This ideal defines the diagonal inside the collision scheme. Its vanishing says that the collision scheme consists only of the diagonal; over $\mathbb{C}$, this is equivalent to polynomial automorphy.

The same construction has a generic field-theoretic form. Write

$$A = \mathbb{C}[x_1, x_2], \quad B = \mathbb{C}[P, Q] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

For a Keller map, L/K is finite and separable, and

$$d_F = [L : K]$$

is the generic degree of F . At the generic point, Paper I gives the compatible isomorphism

$$K \otimes_B C_F \simeq_K L \otimes_K L,$$

with base-changed diagonal evaluation equal to tensor multiplication.

Paper I already uses classical Galois rigidity to exclude generic degrees one and two in every dimension. The purpose here is to realize the quadratic exclusion explicitly in the planar collision ring.

This paper specializes that construction to $d_F = 2$. There are two complementary descriptions of the collision obstruction. The first is geometric and explicit: divided differences produce

a secant determinant and an idempotent p_F generating $\text{Obs}(F)$. The second is generic and field-theoretic: a separable quadratic extension satisfies

$$L \otimes_K L \simeq_L L \times L,$$

and tensor multiplication is projection onto the marked first factor. For a Keller map, the second factor is a genuine residual generic collision, forcing $p_F \neq 0$.

Classical Galois rigidity gives the opposite conclusion. Since every separable quadratic extension is normal, a hypothetical planar Keller map with $d_F = 2$ would be a polynomial automorphism [Cam73, Raz79, Wri81]. Its collision scheme would then be the diagonal, so $p_F = 0$. The contradiction excludes generic degree two.

Section 2 recalls, without reproving, the part of the general construction needed here. Section 3 constructs the planar secant projector. Section 4 proves quadratic collision nonvanishing, and Section 5 combines it with Galois rigidity. Section 6 records the formalization boundary.

This is a generic-degree-two theorem, not a proof of all of $J\mathcal{C}(2)$. The remaining possible counterexamples have nonnormal function-field extensions of generic degree at least three. No normalization, boundary, or inertia construction is needed below.

2 Inputs from Paper I

We use the dimension-independent construction of Paper I [vdV26]. To keep the two manuscripts notationally identical, let

$$F = (F_1, \dots, F_n): \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n,$$

and put

$$\begin{aligned} A &= \mathbb{C}[x_1, \dots, x_n], & S &= \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n], \\ I_R(F) &= (F_i(x) - F_i(y))_{i=1}^n, & I_{\Delta} &= (x_i - y_i)_{i=1}^n, \\ C_F &= S/I_R(F), & R_F &= \text{Spec } C_F. \end{aligned}$$

Since $I_R(F) \subseteq I_{\Delta}$, diagonal substitution descends along the quotient map $q_F: S \twoheadrightarrow C_F$ to

$$\bar{\mu}_F: C_F \twoheadrightarrow A, \quad [s] \longmapsto s(x, x).$$

The image of I_{Δ} is the ideal

$$\text{Obs}(F) := q_F(I_{\Delta}) = I_{\Delta}/I_R(F) \subseteq C_F,$$

which defines the diagonal inside R_F .

The following two results are quoted from Paper I; no proof of the general collision construction is repeated here.

Theorem 2.1 (Diagonal kernel and automorphism detection). *For every complex polynomial map as above,*

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_{\Delta}/I_R(F).$$

Moreover,

$$F \text{ is a polynomial automorphism} \iff I_R(F) = I_{\Delta} \iff \text{Obs}(F) = 0.$$

The implication $\text{Obs}(F) = 0 \Rightarrow F$ is a polynomial automorphism uses Ax–Grothendieck [Ax69]; below we need only the elementary forward implication.

Let

$$B = \mathbb{C}[F_1, \dots, F_n] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A),$$

and let

$$\mu_L: L \otimes_K L \longrightarrow L, \quad \ell \otimes \ell' \longmapsto \ell \ell'$$

be tensor multiplication.

Theorem 2.2 (Generic collision bridge). *If L/K is finite, then*

$$K \otimes_B A \simeq_K L$$

and there is a K -algebra isomorphism

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L$$

which carries the base-changed diagonal evaluation to μ_L . Furthermore, flatness of the localization $B \rightarrow K$ identifies

$$K \otimes_B \text{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F)$$

as K -vector spaces.

These statements combine the fiber-product, generic-base-change, and diagonal-kernel results of Paper I. Their compatibility with μ_L marks which generic factor is the diagonal.

Return now to

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2.$$

For a Keller map, P, Q are algebraically independent and L/K is finite separable. We write

$$d_F := [L : K]$$

for its generic degree. The 2 in $J\mathcal{C}(2)$ denotes ambient dimension; the separate condition $d_F = 2$ concerns the generic fiber. The nonvanishing argument below only uses a finite separable extension of degree two. The Keller condition enters through the nonzero constant Jacobian and, at the end, through Galois rigidity.

3 The explicit planar secant projector

Let

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF = c \in \mathbb{C}^\times.$$

Set

$$S = \mathbb{C}[x_1, x_2, y_1, y_2], \quad \mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

3.1 Ordered divided differences

For $H \in \mathbb{C}[x_1, x_2]$, define

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1},$$

$$D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients lie in S . The ordered telescoping identity is

$$H(x) - H(y) = D_1 H (x_1 - y_1) + D_2 H (x_2 - y_2).$$

Define

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \quad M_F|_{\Delta} = JF.$$

Let

$$\delta_F := \det(M_F) \in S.$$

Proposition 3.1 (Secant annihilation). *The secant determinant satisfies*

$$\delta_F I_{\Delta} \subseteq I_R(F), \quad \delta_F|_{\Delta} = c.$$

Proof. Multiplying $\mathbf{r} = M_F \mathbf{d}$ by the adjugate matrix gives

$$\delta_F \mathbf{d} = \text{adj}(M_F) \mathbf{r}.$$

Both entries on the right belong to $I_R(F)$. Since the entries of $\mathbf{d}$ generate I_{Δ} , the first assertion follows. Restriction to the diagonal gives

$$\delta_F|_{\Delta} = \det(M_F|_{\Delta}) = \det JF = c.$$

□

3.2 The idempotent

Let $\bar{\delta}_F := q_F(\delta_F)$ be the class of δ_F in C_F , and put

$$e_F := c^{-1} \bar{\delta}_F, \quad p_F := 1 - e_F = 1 - c^{-1} \bar{\delta}_F.$$

The element e_F is the diagonal projector, while p_F is the off-diagonal collision projector.

Theorem 3.2 (Planar secant projector). *With the notation above,*

$$e_F^2 = e_F, \quad p_F^2 = p_F, \quad e_F p_F = 0, \quad e_F + p_F = 1.$$

Moreover,

$$\text{Obs}(F) = C_F p_F, \quad \text{Ann}_{C_F}(\text{Obs}(F)) = C_F e_F,$$

and hence

$$C_F \simeq A \times C_F p_F.$$

In particular,

$$p_F = 0 \iff \text{Obs}(F) = 0.$$

Proof. Proposition 3.1 gives

$$\bar{\delta}_F \text{Obs}(F) = 0, \quad \bar{\delta}_F - c \in \text{Obs}(F).$$

Equivalently,

$$e_F \text{Obs}(F) = 0, \quad e_F - 1 \in \text{Obs}(F).$$

Hence $e_F(e_F - 1) = 0$, so $e_F^2 = e_F$; the corresponding identities for $p_F = 1 - e_F$ follow.

The second relation says $p_F \in \text{Obs}(F)$. If $j \in \text{Obs}(F)$, then

$$j = (e_F + p_F)j = p_F j,$$

so $\text{Obs}(F) = C_F p_F$. An element annihilates $C_F p_F$ exactly when its p_F -part is zero, giving

$$\text{Ann}_{C_F}(\text{Obs}(F)) = C_F e_F.$$

Complementary idempotents give

$$C_F \xrightarrow{\sim} C_F e_F \times C_F p_F, \quad a \mapsto (ae_F, ap_F).$$

Diagonal evaluation restricts to an isomorphism $C_F e_F \simeq A$: if $a = \bar{\mu}_F(r)$, then $a = \bar{\mu}_F(re_F)$, while its kernel is $C_F e_F \cap \text{Obs}(F) = 0$. Thus $C_F \simeq A \times C_F p_F$. Finally, $\text{Obs}(F) = C_F p_F$ gives

$$p_F = 0 \iff \text{Obs}(F) = 0.$$

□

4 Quadratic generic collisions

We now work at the generic point. The essential field-theoretic fact is the degree-two marked-root decomposition.

4.1 The quadratic self-tensor product

Let L/K be finite separable of degree two. Paper I's marked-root tensor decomposition applies to a primitive generator α : it splits the marked root from the residual polynomial

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Here h_α has degree one, so its quotient algebra is another copy of L . Thus the general result of Paper I [vdV26] specializes as follows.

Proposition 4.1 (Quadratic marked-root decomposition). *There is an L -algebra isomorphism*

$$\Psi_\alpha: L \otimes_K L \xrightarrow{\sim} L \times L$$

under which tensor multiplication

$$\mu_L: L \otimes_K L \longrightarrow L$$

is first projection. Consequently,

$$\Psi_\alpha(\ker(\mu_L)) = 0 \times L \neq 0.$$

4.2 Nonvanishing of the affine obstruction

Combining the quoted generic collision bridge with Proposition 4.1 transports the nonzero second factor back to the affine obstruction.

Theorem 4.2 (Quadratic collision nonvanishing). *Let $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ be a polynomial map whose induced extension L/K is finite separable of degree two. Then*

$$\text{Obs}(F) \neq 0.$$

Equivalently,

$$I_R(F) \subsetneq I_{\Delta}.$$

Proof. Flatness of $B \rightarrow K$ gives a canonical identification

$$K \otimes_B \text{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F).$$

Under the compatible quadratic product, the right-hand side is

$$\ker(L \times L \xrightarrow{\text{pr}_1} L) = 0 \times L,$$

which is nonzero. Therefore $K \otimes_B \text{Obs}(F) \neq 0$, and hence $\text{Obs}(F) \neq 0$. Since $\text{Obs}(F) = I_{\Delta}/I_R(F)$, this is equivalent to strict containment. $\square$

This theorem is independent of the Keller condition: generic degree two already forces a residual generic collision. For a planar Keller map, the secant projector makes that factor explicit.

Corollary 4.3 (Nonzero planar projector in generic degree two). *Let*

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF = c \in \mathbb{C}^{\times}.$$

If $d_F = 2$, then

$$p_F \neq 0.$$

Proof. The Keller condition makes L/K finite separable. Theorem 4.2 gives $\text{Obs}(F) \neq 0$, and Theorem 3.2 gives $\text{Obs}(F) = C_F p_F$. Therefore $p_F \neq 0$. $\square$

5 Galois rigidity and the collision contradiction

We now use the classical Galois case of the Jacobian conjecture.

Theorem 5.1 (Classical Keller–Galois rigidity). *Let*

$$F: \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2$$

be a Keller map. If the induced finite separable extension L/K is normal, equivalently Galois, then F is a polynomial automorphism.

This is the classical normal/Galois case [Cam73, Raz79, Wri81]; we use it as a literature theorem and do not reprove it here.

Every separable extension of degree two is normal: if the minimal polynomial of a primitive generator has one root in L , its other root also lies in L . Thus Theorem 5.1 applies automatically on the generic-degree-two Keller locus.

Proposition 5.2 (Rigidity forces collision vanishing). *Let*

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF = c \in \mathbb{C}^\times.$$

If $d_F = 2$, then Keller–Galois rigidity gives

$$\text{Obs}(F) = 0, \quad p_F = 0.$$

Proof. The extension L/K is finite separable of degree two, hence Galois. Theorem 5.1 makes F a polynomial automorphism. Theorem 2.1 then gives

$$I_R(F) = I_\Delta, \quad \text{Obs}(F) = 0.$$

Finally, Theorem 3.2 gives $p_F = 0$. □

Theorem 5.3 (Exclusion of generic degree two). *No planar Keller map has generic degree two. Equivalently, every polynomial map*

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF \in \mathbb{C}^\times,$$

satisfies

$$d_F \neq 2.$$

Proof. Suppose that $d_F = 2$. Corollary 4.3, which reads the residual factor in

$$K \otimes_B C_F \simeq_K L \times L,$$

gives $p_F \neq 0$. Proposition 5.2 gives $p_F = 0$. This is a contradiction. □

Equivalently, the two halves of the proof say

$$d_F = 2 \implies \text{Obs}(F) \neq 0$$

by the quadratic generic collision product, while

$$F \text{ Keller and } L/K \text{ normal} \implies \text{Obs}(F) = 0$$

by Galois rigidity and automorphism detection. The secant construction expresses both statements as nonvanishing and vanishing of one canonical idempotent.

6 Lean formalization status

Lean correspondence. The focused theorem spine is formalized in Lean 4 and Mathlib [dMU21, The20]. The general collision objects are defined in `CollisionIdeals/General/Collision`. The function-field localization and generic collision bridge are in `CollisionIdeals/General/GenericFiber/FunctionField.lean` and `CollisionIdeals/General/GenericFiber/CollisionDiagonal.lean`.

The degree-two marked-root calculation is proved in

`CollisionIdeals/General/GenericFiber/QuadraticDecomposition.lean`. Its composition with the polynomial generic-collision bridge, including

$$K \otimes_B C_F \simeq_K L \times L$$

and the descent to $\text{Obs}(F) \neq 0$, is in `CollisionIdeals/General/GenericFiber/QuadraticCollision.lean`.

The coefficientwise divided differences, secant determinant, annihilation relation, diagonal restriction, and canonical projector are proved in `CollisionIdeals/Planar/ExplicitSecant.lean`. The planar nonvanishing and vanishing statements are in `CollisionIdeals/Planar/GenericDegreeTwo/Nonvanishing.lean` and `CollisionIdeals/Planar/GenericDegreeTwo/Vanishing.lean`. Their collision-native contradiction and $d_F \neq 2$ are assembled in `CollisionIdeals/Planar/GenericDegreeTwo/Main.lean`.

The classical theorem in Theorem 5.1 is neither installed as a Lean axiom nor proved in the formalization. Instead, `CollisionIdeals/General/Automorphism/GaloisRigidity.lean` defines the proposition

`ComplexKellerGaloisRigidity 2`

and passes a proof of it explicitly to every theorem using the literature input. Lean therefore proves the degree-two exclusion relative to this single named hypothesis.

No normalization, ramification, boundary, purity, or finite-étale interface occurs in the focused proof. Those constructions belong to the separate problem of nonnormal generic degree at least three.

7 Conclusion

For a planar Keller map, the divided-difference determinant gives

$$C_F \simeq A \times C_F p_F, \quad \text{Obs}(F) = C_F p_F,$$

where p_F is the explicit off-diagonal collision projector. Generic degree two forces

$$K \otimes_B C_F \simeq_K L \times L,$$

with diagonal evaluation equal to first projection, so $p_F \neq 0$.

The quadratic extension is nevertheless Galois. Classical Keller–Galois rigidity would make the map a polynomial automorphism, forcing the collision scheme to equal its diagonal and $p_F = 0$. This incompatibility proves that no planar Keller map has generic degree two.

The remaining possible counterexamples to $J\mathcal{C}(2)$ have nonnormal function-field extensions of generic degree at least three. The degree-two case itself is complete once the classical rigidity theorem is taken as input.

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